

ABIR DAS Associate Software Engineer

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📍 Bengaluru, India

PROFESSIONAL SUMMARY

Experienced Data and AI Engineer with expertise in building scalable data pipelines, maintaining data warehouses, and deploying AI solutions. Proven track record in implementing RAG systems, and orchestrating workflows with Apache Airflow to deliver high-impact AI features and analytics solutions.

PROFESSIONAL EXPERIENCE

RED HAT INDIA PVT LTD

- Associate Software Engineer, DATA & AI**12/2024 – Present | Bengaluru, India
 - Build and optimize scalable data pipelines using Python and SQL for enterprise data processing.
 - Maintain Snowflake data warehouse infrastructure, ingest enterprise sources via Fivetran, design and maintain dimensional data models using dbt, with documentation and tests for analytics and reporting.
 - Orchestrate complex workflows using Apache Airflow (Astro) with DAGs for automated data processing.
 - Built ValidX, a data validation framework (Python, YAML, Snowflake) with multi-env support, CI/CD integration (GitLab/GitHub), Slack alerts, and optional LLM-based failure summarization and root-cause hints.
 - Built an agentic on-call RCA system for data products: LangGraph agent + MCP tools that resolve Atlan assets, sync dbt manifests, and auto-trace column lineage through compiled SQL to root-cause data quality incidents.
- Trainee, DATA & AI**06/2024 – 12/2024
 - Developed end-to-end AI chatbot with UI for training materials using LLMs and RAG architecture.
 - Built complete system with session management, evaluation framework, and VectorDB integration.
 - Deployed on OpenShift for scalability, enabling natural follow-up conversations and reducing navigation time.
 - Achieved 95% user satisfaction rate through efficient similarity search and context retrieval.

EDUCATION

- Master of Operational Research**, University of Delhi2023
- Bachelor of Science in Statistics**, University of Calcutta2020

TECHNICAL SKILLS

- Programming**
Python, SQL, Git, FastAPI
- Data Engineering**
Snowflake, Dbt, Atlan, Fivetran, ETL/ELT
- ML/AI**
LangGraph, MCP, AI Agents
- Cloud/DevOps**
OpenShift, CI/CD
- Orchestration**
Apache Airflow (Astro), DAGs, Workflow Automation
- Visualization**
Streamlit

KEY PROJECTS & ACHIEVEMENTS

- AI Training Chatbot**, End-to-end system with LLMs, RAG, VectorDB, deployed on OpenShift
- GIDS 2023 Speaker**, Presented research on website optimization algorithms

CERTIFICATIONS

- Machine Learning Specialization**
Stanford University (Coursera)
- Data Analytics Professional Certificate**
Google & Kaggle